

Ng Guoyou

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github.com/gyng

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Experience

SPH Media

SPH Media is a leading media organisation in Singapore with around 2500 employees.

Head, Digital Platform, Singapore

Apr 2024–Present

- Managing around 15 reports in multiple teams on application platforms, content platforms (indexing, delivery, creation, commercial products), and AI labs. With the teams and other cross-org initiatives I support all software development in SPH across multiple divisions.
- Developed mid- and long-term strategies, roadmaps, pitches, and mandates for teams.
- Developed and implemented engineering policies, standards, and processes for teams around technical design, development, and operations.
- Guided technical and architecture designs as part of the technical design process.
- Worked on hiring processes and career advancement frameworks.
- Cost-optimised over >\$500K SGD yearly by eliminating unnecessary SaaS contracts, implementing cloud cost optimisation processes, and streamlining teams.
- Prepared for and executed on a smooth Singapore General Elections 2025 which saw a significant spike in site traffic.

Lead Software Engineer, Singapore

Apr 2022–Apr 2024

TypeScript, Node, Python, AWS, React, Vite, Docker, OpenSearch

- Led around 10 developers across two teams to build platform services.
- Architected, developed, and ran an in-house web deployment platform for SSR/ISR Vite/React applications designed for reliability on AWS (think Vercel). The platform implements edge middleware, incremental-static regeneration, a Backend-for-Frontend framework, and deployment workflows. This web platform serves millions of page views per day across > 15 publications.
- Assisted teams with migration away from Drupal and WordPress setups.
- Architected, developed, and ran the content API and image optimisation service. This includes content indexing as a service using OpenSearch (think Algolia), ingestion pipelines, content delivery, and on-the-fly image optimisation service using sharp.
- Architected and developed the in-house browser clickstream tracking script, associated backend service, and session ID tracking scheme.
- Built proof-of-concept research projects including a production-scale liveblog and an edge-based ABAC policy engine.

Basis AI (acquired by Aicadium)

Basis AI was a startup that built a machine learning platform for data science teams.

Software Engineer, Singapore

Jan 2019–Oct 2020, Jun 2021–Apr 2022

TypeScript, Python, Rust, Webpack, React, Redux, Docker, Kubernetes, UX

- Employee #3. Full stack developer on a machine learning platform for training and deployment of data science workloads.
- Main contributor to frontend with Webpack, TypeScript, React+Redux. Created reusable React component library.
- Built client workload monitor in Rust; includes streaming ZIP and large file uploads for workload artifacts.
- Developed backend features in Python such as pagination and RBAC.
- Implemented Docker image caching for fast workloads using kaniko.
- Built company website in Next.js.
- Conducted basic UX research and user testing.

Credit Suisse

Software Engineer, Singapore

Nov 2020–Jun 2021

TypeScript, React, Redux, MobX, Webpack

- Maintained and implemented features for internal financial frontends related to securities order placement, disclosures, and compliance.

GovTech (Government Technology Agency)

GovTech is a Singapore government agency responsible for citizen-facing digital services.

Software Engineer, Singapore

Jan 2016–Jan 2019

TypeScript, React, Redux, Webpack, Python, Rust, Ruby on Rails

- [MyCareersFuture.gov.sg job portal](#), 2018. Orchestrated the codebase transition to TypeScript, Node, and React. Implemented features and bug fixes for the site.
- [OpenCerts.io](#), 2018. Created frontend tools to bulk-register academic transcripts on Ethereum using TypeScript and React.
- Data ingestion validator, API endpoint, and data visualisation application, 2017. Improved data ingestion and validation time by 50 times over existing Python script. Developed frontend application for visualising geographic data from scratch with IE11 support in Flow, React, and Deck.gl.
- [Business Grants Portal](#), 2016. Implemented features for and fixed bugs on forms with complex validation logic. Integrated authentication with SingPass SAML. Introduced linting for Ruby and JavaScript on a large codebase.

Roomorama

Intern, Software Engineer, Singapore

Summer 2013

Ruby on Rails

Education

Bachelor of Computing (Communications & Media), Hons.

National University of Singapore, 2011–2015

UX, HCI, and computer graphics focus

Personal

Singapore citizen. Served 2 years of National Service in the Singapore Armed Forces. Native English and Mandarin Chinese speaker.

Personal Projects

- [A now-playing desktop widget for Windows using Rust, Tauri, and Svelte](#), 2023
- [A zero-config, static, file-based album generator](#), 2022
- [TypeScript library to fetch file listings and individual files from a remote ZIP file](#), 2021
- [Browser extension for organising downloads with >3000 users](#), 2018
- [Retro image filtering application in React and Redux](#), 2017
- [Chatbot for IRC, Slack, Discord in Ruby with a plugin system](#), 2016
- [Toy audio synthesiser library in Rust with MIDI support and WebAssembly interface](#), 2014
- [Sharing sessions and presentations](#)

More projects on [GitHub](#).